1 Solution — GIAM 6.1 Problem 2

1.1 Problem

What is the graph of "$=$" in $\mathbb{R} \times \mathbb{R}$?

1.2 The Graph

The relation "$=$" on $\mathbb{R}$ is, by definition, the set of ordered pairs $(x, y) \in \mathbb{R} \times \mathbb{R}$ whose two coordinates agree:

$$\text{graph}(=) \;=\; \{(x, y) \in \mathbb{R} \times \mathbb{R} \;:\; x = y\} \;=\; \{(x, x) : x \in \mathbb{R}\}.$$

Plotted in the Cartesian plane, this is the line

$$y = x,$$

passing through the origin with slope 1 — the *main diagonal* of $\mathbb{R}^2$. It is commonly denoted

$$\Delta_{\mathbb{R}} \;=\; \{(x, x) : x \in \mathbb{R}\},$$

or simply Δ when the underlying set is clear.

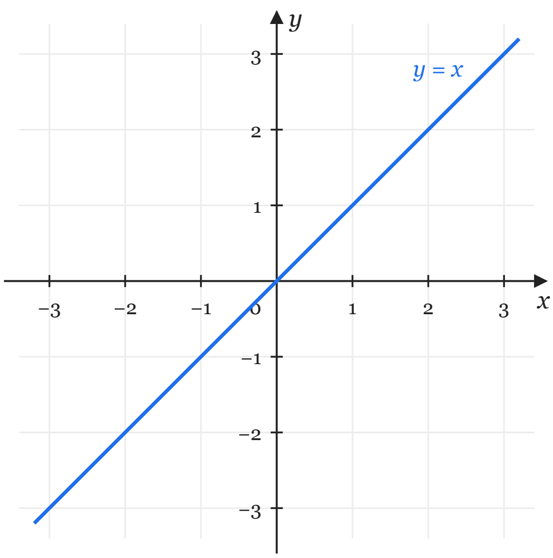

Figure 1.1: The graph of equality on the reals: the line y = x.

1.3 Remarks

Equality is the prototypical equivalence relation, and the diagonal makes its three defining properties visible at a glance:

- **Reflexivity.** Every point on the diagonal has the form (x, x), so $x = x$ for all $x \in \mathbb{R}$.
- **Symmetry.** The set is fixed by the swap $(x, y) \mapsto (y, x)$; the diagonal is its own mirror image across itself.
- **Transitivity.** If (x, y) and (y, z) both lie on Δ, then $x = y = z$, so $(x, z) \in \Delta$ as well.

More generally, the graph of *any* equivalence relation $\sim$ on $\mathbb{R}$ contains the diagonal, since reflexivity forces $(x, x) \in \sim$ for every $x \in \mathbb{R}$.